Algorithms and Optimization
Project Guidelines: Flying Robots with Constraints

April, 16th 2025

1. Project Rules

This project assesses your ability to formulate, implement, and analyze optimization problems involving, for example, control strategies and constraints, similarly to as discussed in class. The specific scope of this project refers to a group of autonomous flying robots navigating through space under physical and operational constraints. Key logistical details of the project are outlined below:

- **Group Composition:** Work in groups of up to **4 students**. Submit your group composition on Moodle by **May 8th**.
- **Deadline:** Submit the final project by **June 8th, 23:59**. Late submissions incur a **-2 points penalty per day** up to a maximum delay of **3 days**.
- **Submission:** Upload your code and a PDF report (expected 5-10 pages) via Moodle.
- **Defense:** Mandatory oral defense on the week of **June 16**. No defense = no grade.

2. Project Description

The main goal of the project is to design and solve an optimization problem that controls a team of autonomous flying robots to perform a coordinated downstream task while minimizing either fuel consumption or total distance traveled. The robots must comply with a set of physical and operational constraints.

Namely, we consider N robots. For each robot $i = 1, 2, \ldots, N$ we define the main variables characterizing the state of the robot:

- $\mathbf{x}_i(k) \in \mathbb{R}^3$: position of robot i at time steps $k = 0, 1, 2, \ldots, n$
- $\mathbf{v}_i(k) \in \mathbb{R}^3$: velocity of robot i at time steps $k = 0, 1, 2, \ldots, n$
- $\mathbf{u}_i(k) \in \mathbb{R}^3$: thrust vector (control input) applied to robot i at time steps $k = 0, 1, 2, \ldots, n$.

These are the optimization variables of the problem that you will formulate. A particular candidate solution to the problem can be abstracted as a vector $\mathbf{Y} \in \mathbb{R}^{9N(n+1)}$ which

collects all the variables listed above. Each robot i contributes $9(n+1)$ variables—corresponding to its position, velocity, and thrust vectors—across all $n+1$ time steps. More concretely, the vector $\mathbf{Y} \in \mathbb{R}^{9N(n+1)}$ collects the positions, velocities and thrust applied over all robots across time.

Given a cost function $\mathcal{C} : \mathbb{R}^{9N(n+1)} \to \mathbb{R}$, we have that $\mathcal{C}(\mathbf{Y})$ quantifies the cost of a candidate solution $\mathbf{Y} \in \mathbb{R}^{9N(n+1)}$. The cost functions to be minimized, along with the constraints, are described next. You will be asked to analyze the convexity of the cost functions and constraint subsets.

2.1 Cost Function: Fuel Consumption

Fuel consumption for each robot is related to the magnitude of the underlying thrust $\mathbf{u}(t)$ exerted at the robot over time t. More concretely, it can be characterized as

$$\mathcal{F}(t) = \int_0^t f(||\mathbf{u}(s)||)\, ds, \tag{1}$$

where $\mathcal{F}(t)$ stands for the fuel consumed (or a proxy of it thereof) up until time t, $f : \mathbb{R}_{\geq 0} \to \mathbb{R}$ is a monotonically increasing convex function, and $||\mathbf{u}(s)||$ refers to the L_2 or Euclidean norm of the vector $\mathbf{u}(s) \in \mathbb{R}^3$ at time s. In other words, $||\mathbf{x}||$ is loosely referred to as the magnitude of the vector $\mathbf{x}$ and it is defined as

$$||\mathbf{x}|| = \sqrt{x_1^2 + x_2^2 + x_3^2}, \tag{2}$$

for a vector $\mathbf{x} = [x_1 \; x_2 \; x_3]^\top \in \mathbb{R}^3$.

In practice, the problem must be addressed in discrete time. Thus, the continuous-time expression in equation (1) is approximated by its discrete-time counterpart:

$$\mathcal{F}(n) = \sum_{k=0}^{n} f\left(||\mathbf{u}(k)||\right) h, \tag{3}$$

where h is the time step and $t = kh$ denotes the discrete time instants, with $k = 0, 1, 2, \ldots, n$, where the thrust is measured. Note that, for brevity, we write $\mathbf{u}(k)$ instead of $\mathbf{u}(kh)$—that is, $\mathbf{u}(k)$ refers to the thrust vector at time $t = kh$.

Therefore, when minimizing fuel consumption $\mathcal{F}(n)$, the cost function becomes

$$\sum_{k=0}^{n} f\left(\|\mathbf{u}(k)\|\right), \tag{4}$$

where we omit the constant scaling factor h, since minimizing the expression in equation (3) is equivalent to minimizing the one in equation (4).

In your Python implementation/formulation, you are free to choose any monotonically increasing convex function f. We suggest using either the identity function $f(x) = x$ or the quadratic function $f(x) = x^2$. For instance, if $f(x) = x^2$, the cost function becomes

$$\sum_{k=0}^{n} \|\mathbf{u}(k)\|^2, \tag{5}$$

as explored in class and in the practical sessions on robot dynamics.

Convexity of the Cost Function

Question. Is the cost function

$$\sum_{k=0}^{n} f\left(\|\mathbf{u}(k)\|\right)$$

convex, given that f is a monotonically increasing convex function? Justify.

2.2 Cost Function: Distance Traveled

Another way to penalize excessive motion is by minimizing the total distance traveled by each robot. This cost may be relevant in missions where motion itself is costly—e.g., mechanical wear, stealth considerations, or reduced radar visibility.

Let $\mathbf{x}(t)$ represent the position of a robot at time t. The continuous-time expression for the cumulative distance traveled up to time t is formally given by:

$$\mathcal{D}(t) = \int_0^t \|\dot{\mathbf{x}}(s)\| \, ds. \tag{6}$$

In discrete time, this becomes:

$$\mathcal{D}(n) = \sum_{k=1}^{n} \|\mathbf{x}(k) - \mathbf{x}(k-1)\|, \tag{7}$$

where we are writing $\mathbf{x}(k)$ instead of $\mathbf{x}(kh)$ as in the previous section. This is the cost function to be minimized.

As with the fuel-based cost, you are free to consider alternative formulations in your Python implementation. For instance, a squared distance version (smoother) is also common:

$$\sum_{k=1}^{n} \|\mathbf{x}(k) - \mathbf{x}(k-1)\|^2. \tag{8}$$

Another possible choice could be

$$\sum_{k=1}^{n} \|\mathbf{x}(k) - \mathbf{x}(k-1)\|_1, \tag{9}$$

where $\|\mathbf{z}\|_1 = \sum_{i=1}^{N} |z_i|$ is the L_1 norm of a vector $\mathbf{z} = [z_1 \ z_2 \ \dots \ z_N]^\top$.

Convexity of Distance-Based Cost

Question. Is the cost function

$$\sum_{k=1}^{n} \|\mathbf{x}(k) - \mathbf{x}(k-1)\|$$

convex? What about the squared version? What about the L_1 norm version?

2.3 Physical Constraint: Newton's Second Law

Each robot must obey Newtonian dynamics in 3D space, subject to gravitational acceleration and a drag force. More concretely, the dynamics of each robot i follow Newton's second law:

$$\mathbf{F}(t) = m\mathbf{a}(t), \tag{10}$$

where m is the mass of the object, $\mathbf{F}(t)$ is the net force applied to the object at time t, and $\mathbf{a}(t)$ is the resulting acceleration. Note that both $\mathbf{F}(t)$ and $\mathbf{a}(t)$ are vector quantities, i.e., $\mathbf{F}(t), \mathbf{a}(t) \in \mathbb{R}^3$. In other words, the net force applied to an object has a direction, and the resulting acceleration is aligned with that direction.

For each flying robot i, we have

$$m_i\, \mathbf{a}_i(t) = m_i \cdot \dot{\mathbf{v}}_i(t), \tag{11}$$

where $\dot{\mathbf{v}}_i(t)$ is the time derivative of the velocity of robot i. On the other hand, the net force applied to robot i is modeled as:

$$\mathbf{F}_i(t) = \underbrace{\mathbf{u}_i(t)}_{\text{Thrust}} - \underbrace{\beta \mathbf{v}_i(t)}_{\text{Drag}} + \underbrace{m_i \cdot \mathbf{g}}_{\text{Gravity}}, \tag{12}$$

where $\mathbf{g} = [0, 0, -9.81]^T$ is the gravitational acceleration vector, β is the drag coefficient, m_i is the mass of robot i, $\mathbf{x}_i(t)$ is its position, and $\mathbf{u}_i(t)$ is the thrust applied at time t.

Therefore, in view of Newton's second law, i.e., via combining equations (11), (12) and (13), we have the continuous-time dynamics for robot i:

$$\boxed{m_i \cdot \dot{\mathbf{v}}_i(t) = \mathbf{u}_i(t) - \beta \cdot \mathbf{v}_i(t) + m_i \cdot \mathbf{g}} \tag{13}$$

Discretization Task

Question 1. Discretize the continuous-time dynamics in equation (14) using a time step of size h, following the approach presented in class.
Question 2. Is this constraint convex? That is, does the set of points satisfying this constraint form a convex subset of the optimization space?
The optimization variables involved in the formulation of this constraint are:

- $\mathbf{x}_i(k) \in \mathbb{R}^3$: position of robot i at time step k
- $\mathbf{v}_i(k) \in \mathbb{R}^3$: velocity of robot i at time step k
- $\mathbf{u}_i(k) \in \mathbb{R}^3$: thrust vector (control input) applied to robot i at time step k.

2.4 Operational Constraint: Wireless Communication

Robots need to be able to communicate throughout the entire navigation period. To ensure this, it is required that no two robots are farther apart than a fixed distance $d > 0$ at any discrete time step.

Communication Constraint

Question 1. Formalize this constraint. The main optimization variable involved is the discrete-time position of the robots, denoted by $\mathbf{x}_i(k) \in \mathbb{R}^3$ across time $k = 0, 1, 2, \ldots, n$.
Question 2. Is this constraint convex? That is, does the set of points satisfying this constraint form a convex subset of the optimization space?

2.5 Operational Constraint: Waypoint Tracking

In many coordinated tasks, it is essential that certain robots pass through specific locations in space at prescribed times. These locations, called *waypoints*, act as milestones that must be reached to ensure mission success. This conforms to a critical component of missions involving distinct tasks—e.g., scanning a target area, dropping supplies, synchronizing with another team, or tracking a fire, to cite a few examples.

Formally, let $\mathbf{w}_{i,\ell} \in \mathbb{R}^3$ denote the ℓ-th waypoint that robot i must pass through, and let $k_{i,\ell}$ denote the corresponding discrete time step at which this waypoint must be reached by robot i.

Waypoint Constraint

Question 1. Formally write the set of constraints enforcing that robot i must pass through each of its assigned waypoints at the corresponding times.
Question 2. Is this constraint convex? That is, does the set of points satisfying this constraint form a convex subset of the optimization space?

2.6 Operational Constraint: Maximum Altitude

To ensure safe and sustainable flight, each robot must remain below a maximum height threshold during its mission. This upper bound reflects both energy limitations (as air becomes thinner at high altitudes, reducing lift and sensor effectiveness) and regulatory airspace constraints—avoid interference with commercial or military air traffic corridors.

Let $h_{\max} > 0$ denote the maximum allowable altitude. This constraint must hold at all time steps for every robot.

Altitude Constraint

Question 1. Express the altitude constraint mathematically. The relevant optimization variable is the robot's position $\mathbf{x}_i(k) = [x_i(k), y_i(k), z_i(k)]^\top \in \mathbb{R}^3$. You may assume the vertical axis corresponds to the z-coordinate.
Question 2. Is this constraint convex? That is, does the set of points satisfying this constraint form a convex subset of the optimization space?

3. Python Implementation

You are now required to implement the formulated optimization problem in Python, including the cost function associated with minimizing fuel consumption and the cost function associated with minimizing the distance traveled along with all operational and

physical constraints previously described. Your implementation should use the CVXPY numerical optimization solver.

3.1 Simulation Parameters

To simulate and solve your optimization problem, you will need to define certain hyper-parameters. You may initially adopt the default values listed below and adjust them as needed for more complex scenarios and discussion:

- **Number of robots:** $N = 5$
- **Time step:** $h = 0.1$
- **Number of time steps:** $T = 50$
- **Robot mass:** $m_i = 1$ (assumed equal for all robots)
- **Drag coefficient:** $\beta = 0.1$
- **Gravitational acceleration:** $\mathbf{g} = [0, 0, -9.81]^\top$
- **Maximum altitude:** $h_{\max} = 10$
- **Communication radius:** $d = 3$

Waypoints. You should also generate the *waypoints*. Ensure that each robot has at least one assigned waypoint. Moreover, simulate an instance in which the waypoints of distinct robots are spaced farther apart than the communication radius d. This creates a meaningful challenge for coordination, as robots must reach their waypoints without violating the distance constraint.

Initial and Final Positions. The initial and final positions of each robot must be on the ground, that is, $z_i(0) = z_i(T) = 0$, where z_i denotes the altitude (third coordinate) of the position $\mathbf{x}_i(k)$ of robot i.

3.2 Output and Visualization

Once your solution is computed, you must generate informative plots to help visualize and validate your results. This will allow you to check whether the constraints are being satisfied and understand the behavior of the system.

You are required to simulate and visualize the system under both the fuel-based and distance-based cost functions. Compare how the choice of cost function influences the resulting trajectories and control behavior.

Simulation Outputs

Question. After solving the optimization problem, return the following plots *for each cost function tested* (i.e., fuel-based and distance-based):

- 3D trajectories of each robot over time.
- Thrust magnitude $\|\mathbf{u}_i(k)\|$ as a function of time for each robot.
- Cumulative fuel consumption $\mathcal{F}_i(k)$ over time for each robot.
- Pairwise distances $\|\mathbf{x}_i(k) - \mathbf{x}_j(k)\|$ over time between all robots.

Make sure to label axes and use legends or color to distinguish different robots. In your report, include relevant comparisons between the results obtained under the different cost functions.

4. Deliverables

Your submission must include two components: i) the complete Python implementation of your optimization problem; and ii) a well-organized concise PDF report. Each group should upload both files via Moodle by the specified deadline.

4.1 Python Code

Submit a single '.zip' file containing all your Python code and simulations. The code must be cleanly structured, documented, and reproducible.

4.2 PDF Report (expected 5 to 10 pages)

The report must be written clearly and concisely and include the following:

- **Discretization and modeling choices:** Justification of the discrete-time model used for dynamics and other constraints.
- **Discussion of questions:** Written answers to all boxed questions throughout the project (e.g., convexity analysis).
- **Cost function comparison:** A comparison of results using the fuel-based and distance-based cost functions (Sections 2.1 and 2.2). Include plots and insights on how the choice of cost affects trajectories, control inputs, and total performance.
- **Plots and results:** All simulation plots required in Section 3.2, with brief explanations and insights.
- **Reflections:** Optional comments on challenges encountered, limitations of your model, or ideas for extensions.

Important: The report should focus on the mathematical formulation, analysis, and interpretation. Do not paste raw code in the report.

4.3 Summary Checklist

Before submitting, make sure your group has completed the following:

✓ Optimization model implemented in Python

✓ Required plots generated and included in the PDF

✓ Mathematical problem formulation clearly written in the report

✓ All boxed questions answered in the report

✓ Simulations performed and discussed for both cost functions (fuel and distance)

✓ Submission uploaded to Moodle before the deadline